Degrees, radians, and angles as rotations

Core Concepts: Degrees, radians, and angles as rotations

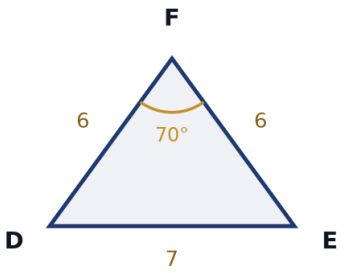
1 In $\triangle ABC$ studying degrees, radians, and angles as rotations, $m\angle A = 35^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 100°

b. 110°

c. 92°

d. 145°



2 A geometric figure scaled by factor $k = 2$ has an original side length of 4 cm. What is the corresponding image length?

a. 6 cm

b. 8 cm

c. 12 cm

d. 16 cm

3 On the coordinate plane, a line with slope $m = 5$ passes through $(0, 9)$. What is its y-coordinate when $x = 1$?

a. 9

b. 16

c. 14

d. 19

Procedural Fluency & Algebraic Manipulation

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4 In $\triangle ABC$ studying degrees, radians, and angles as rotations, $m\angle A = 35^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 93°

b. 103°

c. 85°

d. 145°

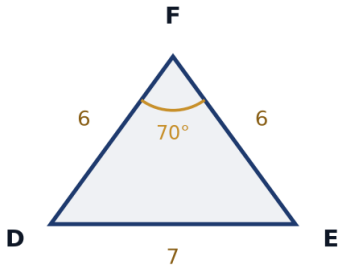
5 A geometric figure scaled by factor $k = 3$ has an original side length of 4 cm. What is the corresponding image length?

a. 9 cm

b. 12 cm

c. 16 cm

d. 36 cm



6 On the coordinate plane, a line with slope $m = 5$ passes through $(0, 9)$. What is its y-coordinate when $x = 2$?

a. 14

b. 21

c. 19

d. 24

Graphical & Geometric Interpretation

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7 In $\triangle ABC$ studying degrees, radians, and angles as rotations, $m\angle A = 35^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 86°

b. 96°

c. 78°

d. 145°

8 A geometric figure scaled by factor $k = 4$ has an original side length of 4 cm. What is the corresponding image length?

a. 12 cm

b. 16 cm

c. 20 cm

d. 64 cm

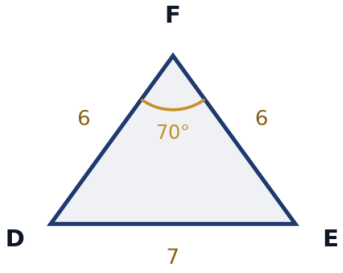
9 On the coordinate plane, a line with slope $m = 5$ passes through $(0, 9)$. What is its y-coordinate when $x = 3$?

a. 19

b. 26

c. 24

d. 29



Exam-Style Problem Solving & Modeling

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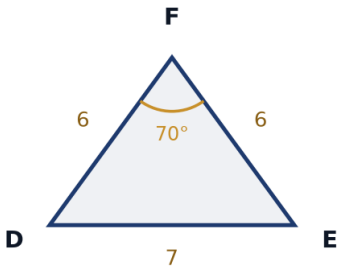
10 In $\triangle ABC$ studying degrees, radians, and angles as rotations, $m\angle A = 35^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 79°

b. 89°

c. 71°

d. 145°



11 A geometric figure scaled by factor $k = 5$ has an original side length of 4 cm. What is the corresponding image length?

a. 15 cm

b. 20 cm

c. 24 cm

d. 100 cm

12 On the coordinate plane, a line with slope $m = 5$ passes through $(0, 9)$. What is its y-coordinate when $x = 4$?

a. 24

b. 31

c. 29

d. 34

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